



4 (a) Define:

(i) stress

.....  
 ..... [1]

(ii) strain.

.....  
 ..... [1]

- (b) Two wires X and Y, with equal unstretched lengths of 0.84 m, are suspended from fixed points that are at the same horizontal level. The lower ends of the wires are attached to a beam of negligible mass. The beam is horizontal and in equilibrium, as shown in Fig. 4.1.

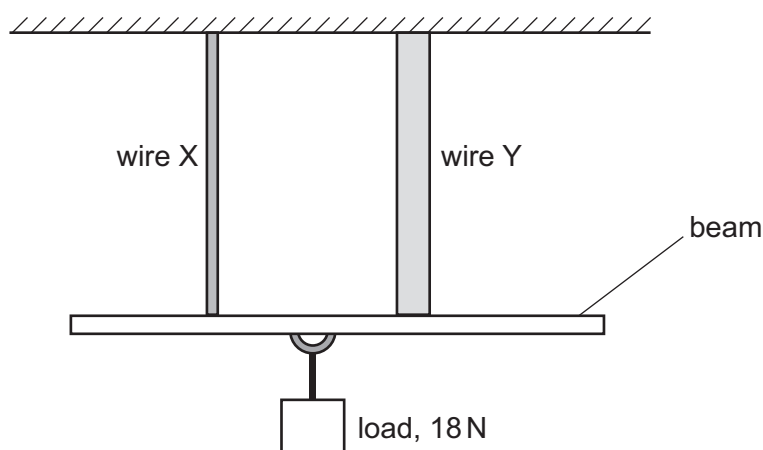


Fig. 4.1

Wire X is made from a metal that has a Young modulus of  $1.9 \times 10^9$  Pa.  
 Wire Y is made from a different metal.

A load of weight 18 N is suspended from the beam at a point that is equidistant from the two wires. This load causes both wires to extend by 0.47 mm.





- (i) Determine the cross-sectional area of wire X.

cross-sectional area = ..... m<sup>2</sup> [3]

- (ii) Wire Y has a greater diameter than wire X.

Explain, without calculation, whether the Young modulus of the metal from which wire Y is made is less than, the same as or greater than  $1.9 \times 10^9$  Pa.

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.....  
.....  
..... [2]

[Total: 7]

